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Mammal seismic line use varies with restoration: Applying habitat restoration to species at risk conservation in a working landscape

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ABSTRACT

Restoration of degraded habitats is increasingly used to mitigate the effects of anthropogenic landscape change on wildlife populations, but wildlife responses to habitat restoration are often assumed rather than verified. In the western Canadian boreal forest, restoration of seismic lines-linear corridors cut for oil exploration-has been proposed to mitigate declines in woodland caribou (*Rangifer tarandus caribou*) driven by altered predator-prey dynamics in industrialized landscapes. Seismic lines fragment caribou habitat, facilitate predator (wolves *Canis lupus* and black bears *Ursus americanus*) movements, and may enhance apparent competition from white-tailed deer (*Odocoileus virginianus)* and moose (*Alces alces)*. Here, we tested the assumption that restoration provides immediate wildlife conservation benefits, using a caribou-boreal forest case study. We used camera traps to monitor seismic line use by caribou, their competitors, and predators following restoration in northeastern Alberta. We assessed species responses to four line strata: two restored (active and passive) and two unrestored (human-use and control). Three-to-six years after restoration, white-tailed deer preferred unrestored seismic lines over actively restored lines, while wolves preferred human-use lines but did not avoid restored lines. Caribou preferred lines in lowland habitat and lines surrounded by low linear density regardless of restoration. Species responses to restoration were muted, indicating that restoration alone may not be immediately effective in stabilizing threatened caribou populations. Our results highlight that wildlife responses to restoration must be tested. We recommend rigorous wildlife monitoring following restoration, particularly when the indirect, interspecific effects of habitat change drive species endangerment or recovery.

1. Introduction

More than 75% of the Earth's terrestrial surface has now been modified by humanity's efforts to acquire resources (Venter et al., 2016). Such drastic landscape transformations trigger widespread responses in wildlife, altering their habitat use, activity patterns, and movement rates (Fisher and Burton, 2018; Wang et al., 2015; Tucker et al., 2018). Ultimately, anthropogenic disturbance is a primary contributor to global biodiversity loss (Maxwell et al., 2016), prompting calls to scale up habitat restoration efforts worldwide (Suding, 2011). However, proposed restoration solutions can be very costly (e.g. Hebblewhite, 2017) and are rarely evaluated for their conservation efficacy (Cooke et al., 2018).

Habitat restoration is the process of assisting the recovery of a disturbed ecosystem; it focuses on returning a disturbed landscape to a reference state by re-introducing or recovering native plant and animal species (McDonald et al., 2016; Shackelford et al., 2013). In doing so, restoration aims to re-establish complex processes that maintain biodiversity, improve ecosystem services, and make the landscape resilient to environmental stress (Brudvig, 2011; Suding et al., 2015). Restoration ecology – which applies the scientific method to restoration – is a relatively recent field that is only beginning to embrace theoretical and evidence-based guidance (Cooke et al., 2018; Shackelford et al., 2013). Outcomes of restoration can be unpredictable and, in the case of wild animals, often go unmeasured entirely (Fraser et al., 2015). All too frequently, site-level restoration is implemented under the assumption that vegetative regeneration will re-establish complex landscape processes, but the assumption is rarely challenged. Crucially, if restoration is aimed at re-building landscapes capable of sustaining wildlife, restoration planning must consider the wildlife community dynamics that the landscape originally supported (Fraser et al., 2015). Without a clear understanding of how restoration efforts impact species-landscape

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relationships, there is little hope of restoration reliably reinstating original landscape functionality.

Western Canadian boreal forests have been transformed into working landscapes where biodiversity competes for space with large-scale industrial development. Oil and gas extraction is a primary driver of boreal landscape change, generating a novel landscape unlike any produced through natural processes (Pickell et al., 2015). Exploration and in situ drilling create a vast network of linear corridors, of which seismic lines are the most common (Lee and Boutin, 2006; Pickell et al., 2015). Seismic lines fragment habitats by increasing the number and decreasing the size of forest patches, while increasing the amount of forest edge (Pattison et al., 2016). They also slow forest succession in this primarily lowland landscape (i.e. dominated by wetland areas, such as bogs and fens), altering abiotic processes at the microsite level that change vegetation successional trajectories (Dabros et al., 2018; Lee and Boutin, 2006). The result of decades of petroleum exploration is an extensive network of "legacy”, or conventional, seismic lines, which are 10–40 years old, 5 − 10 m wide (Dabros et al., 2018) and predicted to remain on the landscape for over 50 years (van Rensen et al., 2015).

Such widespread and persistent networks of linear disturbances affect relationships between species and their landscape, which can in turn have cascading influences on community structure (Fisher and Burton, 2018). Perhaps the most dramatic of these consequences is ongoing and precipitous declines of populations of the woodland caribou (*Rangifer tarandus caribou,*hereafter caribou). Federally listed as threatened in 2002 (COSEWIC, 2002), caribou have declined in part due to changes in predator-prey encounter rates caused by seismic lines, as well as human-mediated apparent competition (McLoughlin et al., 2003; Hervieux et al., 2013). In addition to effects of a warming climate, widespread replacement of mature forest with early seral vegetation has facilitated the northward expansion of white-tailed deer (*Odocoileus virginianus*, hereafter deer; Dawe et al., 2014), which has in turn driven increases in grey wolf (*Canis lupus)*, and subsequently amplified predation pressures on caribou (Latham et al., 2011a). Linear corridors also influence wolf movement by increasing their travel distance and speed across challenging lowland terrain (McKenzie et al., 2012; Dickie et al., 2016) and increasing wolf encounters with caribou (Mumma et al., 2017). Within the boreal forest, caribou have historically mitigated predation risks from wolves by avoiding moose (*Alces alces*) habitat (James et al., 2004). However, the increasingly fragmented landscape has enhanced spatial overlap between moose and caribou and improved wolf access to caribou habitat (Peters et al., 2013; Dyer et al., 2002). Facilitation of wolf movement by seismic lines thus likely increases wolf predation on caribou (Dickie et al., 2016). In addition, black bear (*Ursus americanus)* use of seismic lines could contribute to caribou calf mortality by increasing caribou encounter rates with this opportunistic predator (Latham et al., 2011a,b; Tigner et al., 2014). As caribou prefer lowland bog and fen habitats (James and Stuart-Smith, 2000; Stuart-Smith et al., 1997) where natural regeneration of seismic lines is negligible, these altered predator-prey dynamics are predicted to persist, posing a lasting threat to boreal caribou populations (van Rensen et al., 2015).

Given their widespread footprint and lack of natural regeneration, seismic lines are a high priority target for habitat restoration in western Canada's boreal forests. To return the landscape to its previous functionality (i.e. the ecological processes occurring in the absence of anthropogenic disturbance), restoration should recreate the natural vegetation structure and species on lines (Shackelford et al., 2013; Dabros et al., 2018). At the very least, it should provide resistance to predator movement and facilitate caribou habitat regeneration by reducing early seral vegetation preferred by alternate prey species (Ray, 2014; Bentham and Coupal, 2015). Thus, habitat restoration is considered a promising tool for caribou conservation, but one that has rarely been tested (Bentham and Coupal, 2015).

In this study, we evaluated the assumption that habitat restoration restores the dynamics of animal communities impacted by habitat disturbance. Specifically, we used a large-scale restoration program to test predictions about short-term (i.e. in the years immediately following restoration) responses of mammal species implicated in caribou recovery. We used camera traps (Burton et al., 2015) to monitor seismic line use by caribou, wolves, black bears, white-tailed deer, and moose across four restoration strata: actively restored lines (“active”), passively restored lines (“passive”), human-use lines (“human-use”), and unrestored control lines (“control”) (Fig. 1, Fig. A1, and Appendix A1). We hypothesized that if seismic line restoration is to be effective for caribou conservation, it should re-establish boreal landscape functionality by spatially segregating caribou from their predators and apparent competitors. That is, relative to control lines, we predicted that restored

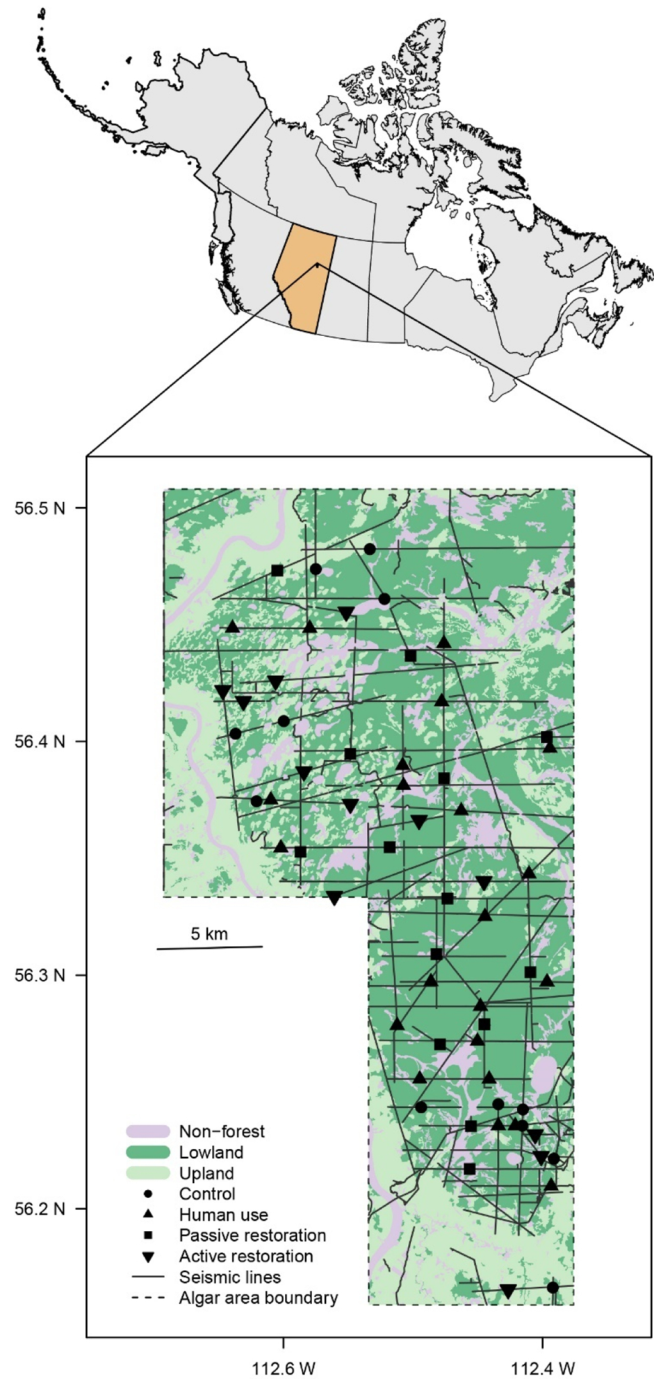

Fig. 1. The Algar study area, camera trap locations, and seismic line strata for the Algar Caribou Habitat Restoration Program located along the east side of the Athabasca River in northeastern Alberta, Canada (56.2588 N, 112.6909 W).

seismic lines would be used less frequently by wolves, black bears, deer, and moose, and used more frequently by caribou. Additionally, we predicted that snow, natural landcover, landscape-scale linear density, line width, and vegetation height would also affect line use by the focal species (Dickie et al., 2017; Fisher and Burton, 2018; Tigner et al., 2014). This study is one of the first to evaluate the efficacy of boreal habitat restoration as a strategy for wildlife conservation (Fraser et al., 2015), and our results may inform restoration practice for re-establishing landscape functionality in support of species at risk conservation across forest ecosystems.

2. Methods

2.1. Study area

Our study took place in the Algar sub-range of the East Side Athabasca River (ESAR) caribou range, approximately 70 km southwest of Fort McMurray, Alberta (56.2588 N, 112.6909 W; Fig. 1). The 570 km^2 area is within the western sedimentary basin of the Canadian boreal forest, and contains vegetation cover broadly typical of this region: lowland habitat of mature black spruce (*Picea mariana)* and tamarack (*Larix laricina)* wetlands, with upland habitat of aspen (*Populus tremuloides)*, white spruce (*Picea glauca)*, and jack pine (*Pinus banksiana)*. We considered this area to have low-intensity disturbance relative to other landscapes in the oil sands region (e.g. Fisher and Burton, 2018): anthropogenic features were almost exclusively legacy seismic, with a total of 524 km of seismic lines and a linear feature density of 1.1 km/km^2 (Fig. 1; Alberta Biodiversity Monitoring Institute, www.abmi.ca). In comparison, Lee and Boutin (2006) reported linear densities as high as 10 km/km^2 in other townships of the oil sands region. Due to the relative lack of other disturbances, we considered additional anthropogenic effects to be negligible compared to seismic lines and did not include them in our analyses.

2.2. Restoration and camera trapping

The Algar Caribou Habitat Restoration Program began in 2011, with a goal of restoring 264 of the 830 ha of legacy seismic lines within the study area (OSLI (Oil Sands Leadership Initiative), 2012; Silvacom and Nexen, 2015). This goal was based on achieving a minimum target of 65% undisturbed habitat within boreal caribou range (Environment Canada, 2012). The program's objectives aimed to address both structural and functional restoration (Dabros et al., 2018) by simultaneously promoting vegetative regeneration and providing movement barriers for caribou predators. We followed internationally accepted terms to define the program's restoration activities (McDonald et al., 2016); specifically, we considered restoration to be any activity with the aim of achieving ecosystem recovery to the reference state, including (i) passive restoration as protection to allow natural (spontaneous) vegetation regeneration, and (ii) active restoration as assisted regeneration and reconstruction. Seismic lines within the study area were assessed and categorized by the program prior to restoration (OSLI (Oil Sands Leadership Initiative), 2012). Line segments considered for active restoration had to meet the following criteria: not used for industrial or trapper/outfitter access, little or no natural regeneration, and accessible for restoration activities. Active restoration included site preparation with mounding and addition of coarse woody material (i.e. dead and damaged trees) where available, as well as planting of black and white spruce seedlings in densities ranging from 400 to 1200 stems per ha. Line segments designated for passive restoration (termed natural regeneration protection by the program) were defined as those with regeneration ≥1.5 m in height (roughly representing a caribou sightline) and crown cover > 50% (OSLI (Oil Sands Leadership Initiative), 2012). For the remaining seismic lines in the study area, we categorized two unrestored strata as "human-use" – line segments left open for industrial or trapper access – and "control" – line segments meeting the conditions for active restoration but set aside as untreated controls for this wildlife assessment (representing a "business as usual" case of no restoration; Fig. A1).

We randomly selected line segments from within each stratum for camera trap sampling locations, deploying a total of 60 Reconyx HyperFire PC900 camera traps (Reconyx, Holman, WI) distributed across strata as follows: 22 active, 12 passive, 14 human-use, and 12 control camera stations (Appendix A1). Camera trap data used in this study were collected between November 2015 and November 2018. During the April and November 2017 site visits we measured site-level characteristics for inclusion as predictor variables (Table A1, described in Appendix A3). Species identifications were made from camera images by two observers (ET, JB) using *Camelot* software (Hendry and Mann, 2017; https://gitlab.com/camelot-project/camelot); any uncertain identifications were referred to a third observer (ACB) and excluded from analysis if they could not be identified with certainty. We used *camtrapR* (Niedballa et al., 2016) within R statistical software (R Core Team 2018; www.r-project.org) to organize image data for statistical analysis. All methods for wildlife monitoring were approved by the Canadian Council of Animal Care administered by the University of British Columbia (protocol A17-0035).

2.3. Data analysis

2.3.1. Model variables

We assessed mammal responses to seismic lines in terms of the number of detection events for each focal species at each camera station in each month (i.e. station-month was our spatiotemporal unit of sampling, Appendix A2). We summarized detection events for each species in *camtrapR* record tables, treating events as independent when occurring at least 30 min apart (Rovero and Zimmermann, 2016). We discretized continuous camera data into month-long period's *sensu* Fisher and Burton (2018). As our statistical sampling unit was "month" and months sample a seismic line multiple times, our sampling units were not independent from one another. We therefore modeled monthly detections within a zero-inflated generalized linear mixed modelling (GLMM) framework, modelling monthly species' detections as a function of seismic line strata, as well as site- and landscape-scale variables hypothesized to affect species' line use (Table A1, Appendix A3). We included a random intercept for station to account for non-independence of monthly detections at the same site. We also included a random intercept of month because we sampled the same calendar month up to three times across the multi-year survey (Appendix A2). For example, observations in January of each year may be similar due to their occurrence in mid-winter. We included the full 36-month sampling period for caribou, deer, moose, and wolves. We only used months between April and October for black bear detections to account for the bears' winter denning period. During the 36-month survey period, camera failures resulted in some inactive camera periods (Table B1), which we accounted for by including the number of active days per month per camera as a model variable.

Our hypothesis testing focused on the role of seismic line strata in explaining variation in the detections of focal species during each station-month. We used the control lines as the reference stratum to represent a "business as usual" case of no restoration. We sought to model additional heterogeneity in mammal line use by including predictor variables for proportion of lowland habitat, snow presence, seismic line density, vegetation height on-line, and line width at each station (Table A1, Appendix A3). For spatial variables – lowland habitat and line density – we performed a multi-scale analysis to determine scales at which habitat best described occurrence for each species (Fig. A2, Appendix A4; Fisher et al., 2011).

2.3.2. Model building

We constructed sets of candidate models with different combinations of predictor variables. We used model selection based on Akaike

Information Criterion (AIC) to assess the relative degree of support among candidate models (Burnham and Anderson, 2002). The null model included only active days − the measure of sampling effort that was included in all models (Table A1). We included restoration stratum in all other candidate models. Our fully parameterized model included all five additional predictor variables (Table A1). As very little is known about species-habitat associations from this region of the boreal forest, we did not have *a priori* reasons to test only specific candidate models. We therefore constructed multiple permutations of candidate models using the "dredge" function from the R package *MuMin* (Bartoń, 2009). This function generated candidate models using combinations of five predictor variables while keeping the strata and active days variables fixed in all candidate models, then performed AIC model selection on the model set (Bartoń, 2009). We considered predictors contained in all models that ranked within 2 ΔAIC of the top model to be influential variables for that species (Burnham and Anderson, 2002). We then compared effect sizes of predictor variables using parameter estimates from a fully parameterized model. We report results as regression coefficients (β) and standard error (SE) for each model variable.

3. Results

Total sampling effort over the 36-month study period was 43 361 camera-days, with a median of 735 active days per camera (range = 326–1108). There were 3 058 independent detection events of mammals recorded (Table B1), from which we identified 16 mammal species (Table B2). Of the 2 309 detections of the five target species, deer were most frequently detected (n = 835) and caribou were least frequently detected (n = 271; Table B2). Seismic line strata was the best-supported predictor variable only for wolves, whereas line use by all other species was more strongly influenced by habitat variables (Fig. 2).

Deer use of actively restored lines was lower than their use of control lines (-1.07 ± 0.418, $p = 0.011$; Fig. 2c), but caribou, predators and moose showed no response to active restoration. Wolf use of human-use lines was higher than of control lines (1.15 ± 0.530, $p = 0.030$; Fig. 2a), but this effect was not seen for black bears, caribou, or caribou competitors. Passive restoration did not have an effect on line use by any of the five target species (Fig. 2).

Lowland habitat had the strongest effect for caribou and deer, with more frequent caribou use (3.47 ± 0.583, $p < 0.001$; Fig. 2e) and less frequent deer use of lowland sites (-1.85 ± 0.354, $p < 0.001$; Fig. 2c). Higher seismic line density resulted in less frequent line use for caribou (-1.32 ± 0.394, $p < 0.001$) but more frequent line use for deer (0.748 ± 0.306, $p = 0.015$). The presence of snow on lines resulted in less frequent line use for caribou and wolves (-1.59 ± 0.291, $p < 0.001$ and -0.608 ± 0.243, $p = 0.012$, respectively). Black bears, deer, and moose line use was higher on lines with taller vegetation (0.677 ± 0.321, $p = 0.035$; 1.11 ± 0.300, $p < 0.001$; and 0.835 ± 0.352, $p = 0.018$; Fig. 2b, c, and d respectively).

For all species except black bear, the model with the highest AIC weight included predictor variables with significant effects on line use (Table 1). The null model had the most support for black bear, indicating that predictor effects were weak. All predictor variables appeared in the top model set for wolves, though line density, vegetation height and line width only occurred in one model each. The top model for deer included lowland habitat, linear density, vegetation height and snow, though line width was also included twice in the three next-best models. Moose line use was best explained by a model containing vegetation height. Finally, the best-supported model for caribou included snow presence, lowland habitat and linear density. In the subsequent two ranked models for caribou, vegetation height and line width each appeared once.

4. Discussion

Seismic line restoration, or lack thereof, influenced line use for only two of the five large mammal species that play a central role in caribou dynamics in northern boreal forests. If linear restoration is to support recovery of threatened caribou populations, it should "remove" the ecological effects of seismic lines by re-establishing spatial segregation between caribou, their predators, and their apparent competitors. However, in the short-term (three-to-six year) period post-restoration covered by this study, we found that only white-tailed deer showed the predicted lower use of restored seismic lines. Contrary to our prediction, caribou predators did not show lower use of lines actively restored via mounding and re-planting, indicating that these treatments did not create movement barriers. However, wolves did show higher use of lines left open for human access (relative to control lines). Caribou used seismic lines in lowland habitat, regardless of the seismic line type. Our results suggest that, at least in the short term, seismic line restoration alone may not re-establish landscape functionality enough to support caribou conservation on the working landscape. Our study challenges the common, yet rarely tested, assumption that habitat restoration aimed at vegetation regeneration supports wildlife conservation on disturbed landscapes, thus emphasizing the need to monitor the outcomes of restoration for wildlife communities.

4.1. Active restoration does not impede predator movement

Contrary to a key hypothesis motivating linear restoration in caribou habitats, wolves and black bears did not use seismic lines less often following restoration by site preparation and planting. Although mounding and application of coarse woody material has been proposed as a barrier to predator movement, providing functional restoration in addition to structural restoration (Bentham and Coupal, 2015; Dabros et al., 2018), our results do not support this assertion. Nevertheless, it is possible that this result is specific to our study area. In the Algar restoration program, coarse woody material could only be applied to lines where it was readily available in the surrounding forest. At many lowland sites, the restoration program reported that there was insufficient material to reach target application for movement barriers. One portion of the study area designated for restoration received 40% of targeted coarse woody material volume, while two other portions received only 16% (OSLI (Oil Sands Leadership Initiative), 2012; Silvacom and Nexen, 2015). In previous studies, line-blocking via tree-felling (where trees were felled every 10 − 15 m along a seismic line) did not impede wolf line use (Neufeld, 2006), but intense application of log debris (approximately 200 m^3/ha over a 200 m line segment) decreased wolf detections relative to open seismic lines (Keim et al., 2019). Taken together, these results suggest that without adequate woody material, mounding alone may not be an effective inhibitor of predator movement.

Consistent with previous studies, we observed higher use of human-use lines by wolves (Dickie et al., 2017). As accessible travel routes, human-use lines in our study experienced more vehicle traffic than other lines (e.g. snowmobile, all-terrain vehicle), causing soil compaction, altered vegetation structure, and decreased likelihood of natural regeneration (Bentham and Coupal, 2015; Dabros et al., 2018; Pigeon et al., 2016). These characteristics of human-use lines, relative to those of the other line types, likely ease movement barriers and permit faster, more efficient travel for wolves (Dickie et al., 2017; Finnegan et al., 2018a). During winter, vehicle traffic compacts snow on lines to provide firmer trails, decreasing energetic travel costs for wolves when snow presence on lines otherwise acts as a deterrent (Droghini and Boutin, 2017).

4.2. Small effects of restoration on caribou line use

Although one of the strategic goals of seismic line restoration is

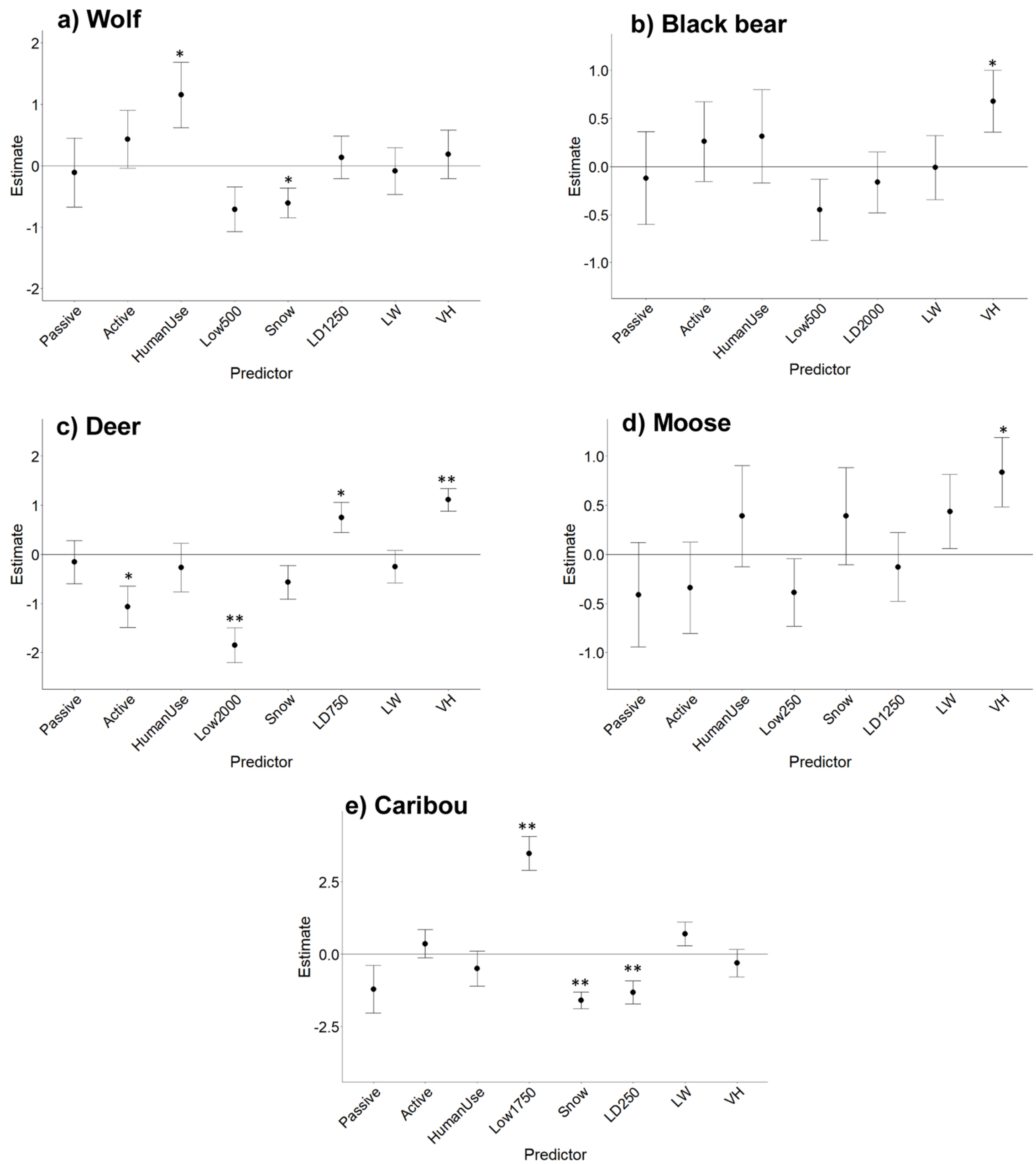

Fig. 2. Estimated effects of seismic line strata and other factors on line use by boreal mammals in northeastern Alberta. Parameter estimates from fully-parameterized zero-inflated GLMMs of monthly detections from 60 camera traps sampled Nov. 2015 – Nov. 2018 are displayed as mean ± standard error. We used control seismic lines as our reference strata to assess the relative effect of other seismic line strata. We centred and standardized predictor variables to indicate effect sizes relative to the control strata, with all other variables held constant at a mean value. We have included additional predictor variable details in Table A1. ** denotes estimates with $p < 0.01$, * denotes estimates with $p < 0.05$.

habitat regeneration that facilitates caribou use and hence conservation, in our study caribou did not use actively restored lines significantly more than controls. This is perhaps not surprising, given the slow regeneration time of native vegetation in this boreal landscape (van Rensen et al., 2015). Instead, caribou use of seismic lines was more strongly influenced by surrounding habitat, season, and landscape-scale linear density than by restoration. Caribou preference for lowlands has been well documented and is a primary motivation for restoration in this habitat (e.g. James and Stuart-Smith, 2000; Latham et al., 2011b). The negative effect of snow on caribou line use indicates decreased line use in winter, likely due to the higher energetic cost of moving through snow accumulated on lines (Fancy and White, 1987).

4.3. Apparent competitor line use

Our prediction that caribou apparent competitors would use restored lines less than controls was supported for deer, but not for moose. Deer used actively restored lines less than control lines, suggesting that active restoration via site preparation and planting is an effective tool for reducing line use for this range-expanding species (Dawe et al., 2014; Latham et al., 2011a). Although we did not observe a strong effect of restoration on moose, their positive responses to vegetation height and line width are consistent with studies reporting moose selection for woody shrubs and open-canopied habitat patches (Latombe et al., 2014; Wasser et al., 2011). These results highlight that functional restoration without structural restoration – that is, creating movement barriers without restoring vegetation composition – is

Table 1
Model selection results for zero-inflated generalized linear mixed models describing effects of seismic line restoration and line characteristicsfor five target mammal species. Top models (those within 2 ΔAIC) indicate which predictor variables (in addition to seismic line strata and active days) best explained seismic line use for each species. Strata = seismic line type, AD = proportion of active days, snow = proportion of snow days, low = lowland habitat, LD = line density, VH = vegetation height, and LW = line width. Values following lowland and line density indicate the buffer radius used to measure these variables. Distribution represents the underlying distribution of the response variable (Appendix A2), df is the degrees of freedom in the model, ΔAIC indicates the difference in AIC scores from the top model, AICwt is the AIC weight attributed to that model, Cum.Wt is the cumulative model weight, and ER is the evidence ratio (AIC weight of top model/AIC weight of model). We have included additional details on predictor variables and modelling approach in Table A1 and Appendices A2 and A3.

Species	Predictor Variables	Distribution	Df	ΔAIC	AICwt	Cum.Wt	ER
Wolf	Strata + AD + Snow + Low500	Nbinom2	11	0.00	0.188	0.188	
	Strata + AD + Snow + Low500 + LD1250		12	1.17	0.104	0.292	1.80
	Strata + AD + Snow		10	1.34	0.096	0.389	1.95
	Strata + AD + Snow + Low500 + VH		12	1.63	0.083	0.472	2.26
	Strata + AD + Snow + Low500 + LW		12	1.81	0.076	0.548	2.48
Black bear	AD	Nbinom2	6	0.00	0.354	0.354	
	Strata + AD + Low500 + VH		11	1.66	0.154	0.508	2.29
White-tailed deer	Strata + AD + Low2000 + LD750 + VH + Snow	Nbinom1	13	0.00	0.324	0.324	
	Strata + AD + Low2000 + LD750 + VH		12	0.53	0.249	0.574	1.30
	Strata + AD + Low2000 + LD750 + VH + Snow + LW		14	1.41	0.160	0.734	2.02
	Strata + AD + Low2000 + LD750 + VH + LW		13	1.97	0.121	0.856	2.68
Moose	Strata + AD + VH	Nbinom2	10	0.00	0.145	0.145	
	Strata + AD + VH + Low250		11	0.490	0.114	0.259	1.28
	Strata + AD + VH + LW		11	0.528	0.112	0.371	1.30
	Strata + AD + VH + LW + Low250		12	1.315	0.075	0.446	1.93
	Strata + AD + VH + Snow		11	1.360	0.074	0.519	1.97
	Strata + AD + VH + Low250 + Snow		12	1.851	0.058	0.577	2.52
	Strata + AD + VH + LW + Snow		12	1.867	0.057	0.634	2.54
	Strata + AD + VH + LD1250		11	1.984	0.054	0.688	2.70
Caribou	Strata + AD + Low1750 + Snow + LD250	Nbinom1	12	0.00	0.326	0.326	
	Strata + AD + Low1750 + Snow + LD250 + LW		13	0.71	0.229	0.554	1.42
	Strata + AD + Low1750 + Snow + LD250 + VH		13	2.00	0.120	0.674	2.71

insufficient as it does not deter apparent competitors from using seismic lines (Ray, 2014; Finnegan et al., 2018a,b).

White-tailed deer expansion into the boreal forest is in part attributed to industrial development (Dawe et al., 2014), as anthropogenic land conversion alters habitat in favour of deer. Deer selection for linear features and cut blocks has been previously documented (Fisher and Burton, 2018), and this relationship is further supported by the positive association observed in our study between deer, line density, and vegetation height. Lee and Boutin (2006) found that unrestored lines retained early seral vegetation long after the initial disturbance, and Finnegan et al. (2018b) documented more important forage species for deer and moose on seismic lines relative to interior forest plots. Decreased early seral forage on restored lines, combined with movement barriers from mounding, could explain the negative responses of deer to actively restored lines observed in our study. Active restoration may therefore provide structural restoration by returning disturbances to natural vegetation faster (Bentham and Coupal, 2015) as well as provide some functional restoration of habitat by deterring a key caribou competitor. Both responses could decrease competitive pressures on caribou over time.

4.4. Directions for future research

We suggest that there are several important implications of our study, as well as several key directions for future research. One such direction would be to expand on our study through inclusion of an additional sampling stratum in undisturbed habitat (“off-line”). Off-line sites could be used to sample species' space use away from local anthropogenic disturbances, thus providing a potential reference target with which to assess the effectiveness of restoration at achieving ecosystem recovery. However, we were limited in our ability to deploy camera traps in interior forest patches within our study area. The largest areas without seismic lines were open fens in lowland habitat (Fig. 1), which were not only difficult to access but would have confounded comparisons with our other sampling strata due to the strong habitat differences. Potential zones of influence around anthropogenic features – considered as a 500 m buffer by Environment Canada (2012) – also made it difficult to set off-line cameras in higher linear density areas within our sampled landscape, as off-line areas would have been close to other lines and thus may not have reliably represented species' presence in undisturbed habitat. Nevertheless, incorporating an “undisturbed” stratum would build on our comparison across line types, and would also help in characterizing population-level responses across the landscape (e.g. Burgar et al., 2019). The ultimate measure of restoration effectiveness for caribou recovery is the stabilization or growth of caribou populations. While the behavioural responses we documented in this study can be linked to population consequences, we recommend complementary monitoring at both behavioural and population scales. Camera traps are a promising tool for monitoring across multiple ecological scales, particularly if landscape-level interventions can be replicated (and monitored) across multiple landscapes and the populations they contain.

We suggest that active restoration in our study system provided few direct benefits to caribou over the short term. However, we caution that ours was a preliminary study conducted shortly after implementation of restoration, and restoration of landscape functionality is a goal usually realized over the longer term (Shackelford et al., 2013). Still, less frequent use of actively restored lines by deer may indicate the beginning of the restoration process. By deterring seismic line use by deer, active restoration may re-establish spatial segregation between caribou and a key apparent competitor, thus reducing predation risk for caribou on restored lines over time. Although caribou predators still used restored lines in our study, use of these lines may decrease as fewer of their prey species are found at those sites. We recommend that future studies should continue to monitor mammal responses over longer periods to allow for greater vegetation regeneration. Further research should seek to relate seismic line restoration to animal movement rates and behaviour in order to assess specific mechanisms behind line use (e.g. Dickie et al., 2017; Finnegan et al., 2018a). We also recommend comparisons of our results with studies in other areas to observe restoration effects across landscapes. Ultimately, responses to restoration at the behavioural level must also translate into population-scale responses in caribou and their interacting species, thereby inducing changes in community-wide landscape use and re-establishing landscape function.

In its latest caribou range plan, the Government of Alberta advocated for an integrated land management approach to conservation, which may include prioritizing restoration in preferred caribou habitat while consolidating industry access routes to reduce human influence on the landscape (Government of Alberta, 2017; Yemshanov et al., 2019). We suggest that our results provide some support for a strategy in which linear restoration is implemented in areas where caribou and deer occurrences overlap to re-establish spatial segregation between caribou and their competitors. However, we call for further research to explore which scale of application would best balance human access, caribou habitat needs and predator movement. Integrated land management may allow for wildlife coexistence alongside industry, but only if sufficient functional habitat is available. Finding a balance between industry access and habitat protection should be a priority in future studies. In addition, a more comprehensive view should account for the cumulative impact of all anthropogenic disturbances on the landscape (Fisher and Burton, 2018; Neilson and Boutin, 2017), and how strategic habitat restoration might mitigate effects on the boreal mammal community.

The United Nations recently declared 2021–2030 the Decade on Ecosystem Restoration, increasing the global emphasis on restoration of impacted landscapes (United Nations Environment, 2019). This focus will require an associated emphasis on monitoring systems to ensure effectiveness of restoration efforts. While restoration monitoring is often focused on vegetation and soils, achieving successful restoration of broader ecosystem functions will require monitoring of vertebrate community responses (Fraser et al., 2015), as we have done here using camera traps. This tool provides the unique opportunity to assess community-wide influences of anthropogenic disturbance and associated mitigations (Fisher and Burton, 2018), which should ultimately improve conservation efficacy. Rather than studying single species, or a single predator-prey interaction in isolation, monitoring multiple species considers the broader ecological context within which focal species fit (Burgar et al., 2019). Multi-species studies are useful for identifying patterns and assessing general effects of landscape change, exposing both direct and indirect relationships between community members (Caravaggi et al., 2017). Not only is this approach valuable for taking a bigger picture approach to species conservation, but it also sheds light on interspecific relationships that have thus far been overlooked.

5. Conclusion

Habitat restoration is a key component of wildlife conservation, but animal responses need to be monitored to ensure the efficacy of restoration (Fraser et al., 2015; Dabros et al., 2018). At the early stage of vegetation recovery in our study, large boreal mammals are showing weak evidence of response to successful habitat restoration. However, continued monitoring is necessary to determine if current trends prevail and ultimately lead to positive change for caribou populations. Boreal habitat regeneration is a long-term goal; therefore, coordinated conservation strategies may need to be applied to sustain herds in the interim (Serrouya et al., 2019). By monitoring such efforts, management actions can be adapted to optimize their effectiveness. Our study is an example of how restoration projects can be monitored for their effects on specific focal species as well as the broader community in which they are embedded. Continued assessment of mammal responses to land management informs how wildlife and industry might best coexist, thereby helping to balance conscientious human development with healthy and sustainable ecosystems.

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Declaration of Competing Interest

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.biocon.2019.108295.

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